Mathematics for AI 2 Exam 2023

1. (a) Write down an antiderivative for the function $g(x) = x^2 + 2x + 3$.

   (b) Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = xe^x$.

      i. Write down an antiderivative F for the function f.

      ii. Write down the antiderivative F for the function f which satisfies the condition $F(0) = 2$.

   (c) Calculate the following integral:

$$\int_0^{\sqrt{\frac{\pi}{2}}} x \cos\left(x^2 - \frac{\pi}{2}\right) \, dx.$$

2. (a) Let $I = (a, b)$, $x_0 \in I$, and $f : I \to \mathbb{R}$. Suppose that f is differentiable at x_0. Give the definition of the derivative $f'(x_0)$.

   (b) Let $n \in \mathbb{N}$, let $I = (a, b)$ and let $f : I \to \mathbb{R}$ be given by $f(x) = x^n$. Use the definition from part (a) of this question to prove that, for all $x \in I$,

$$f'(x) = nx^{n-1}.$$

   (c) Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = e^{x^3 - 12x}$.

      i. Calculate $f'(x)$.

      ii. Write down all stationary points of f.

      iii. Use the second derivative test to determine whether these stationary points are local maxima or minima, if possible.

3. (a) Let $(a_n)_{n\in\mathbb{N}}$ be a complex-valued sequence. Give the definition of what it means to say that $(a_n)_{n\in\mathbb{N}}$ is **bounded**.

   (b) Let $(a_n)_{n\in\mathbb{N}}$ be a complex-valued sequence and $a \in \mathbb{C}$. Give the definition of what it means to say that $(a_n)_{n\in\mathbb{N}}$ **converges** to a.

   (c) Let $(a_n)_{n\in\mathbb{N}}$ be a convergent complex-valued sequence. Prove that $(a_n)_{n\in\mathbb{N}}$ is bounded.

   (d) Give an example of a real-valued sequence $(a_n)_{n\in\mathbb{N}}$ such that

$$\lim_{n\to\infty} a_n = \infty.$$

   (e) Let $a, b > 0$ be fixed. Use the Sandwich Rule to show that

$$\lim_{n\to\infty} \sqrt[n]{a^n + b^n} = \max\{a, b\}.$$

   You may also use, without proof, the fact that $\lim_{n\to\infty} \sqrt[n]{2} = 1$.

4. (a) Let $D \subset \mathbb{R}$, $x_0 \in D$ and $f : D \to \mathbb{R}$. Complete the following definition. We call f **continuous at** x_0 if and only if, for all sequences $(x_n)_{n\in\mathbb{N}}$ in D such that $x_n \to x_0 \ldots$

   (b) Give an example of a function $f : [-1, 1] \to \mathbb{R}$ which is not continuous at 0.

   (c) Let $D \subset \mathbb{R}$, $x_0 \in D$ and $f : D \to \mathbb{R}$. Using the definition from part (a) of this question, prove that f is continuous at x_0 if and only if

$$\forall \epsilon > 0, \exists \delta > 0 : |x_0 - x| < \delta \,\text{and}\, x \in D \Rightarrow |f(x_0) - f(x)| < \epsilon.$$

   (d) Use the result from part (c) of this question to show that the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is continuous at $x_0 = 1$.